THERMODYNAMICS

II B.TECH I Semester : MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME14	PCC	3	1	0	4	40	60	100

COURSE OVERVIEW:

This course intended to introduce thermodynamic laws and their application to analyse boilers, heat pumps, heat engines etc., properties of mixture of gases and pure substances, Psychrometry.

COURSE OBJECTIVES:

The course should enable the students to:

- To understand the basic concepts of thermodynamics.
- To comprehend laws of thermodynamics and apply it to the related processes.
- Perform thermal analysis on behaviour and performance of systems.

COURSE OUTCOMES:

- CO1 - Apply the first law of thermodynamics for simple open and closed systems under steady and unsteady conditions.
- CO2 – Analyse the second law of thermodynamics to open and closed systems and calculate entropy.
- CO3 - Summarize the thermodynamic properties of gases and steam and apply it to related system analysis.
- CO4 - Calculate the properties of gas mixtures and apply it to related system analysis.
- CO5 - Evaluate the properties of moist air and its use in psychometric processes.

UNIT-I Basic Concepts and First Law

Basic concepts – concept of continuum, microscopic and macroscopic approach. Path and point functions. System and their types. Thermodynamic Equilibrium State, Quasi-static, reversible and irreversible processes. Heat and work transfer.

Zeroth law of thermodynamics –First law of thermodynamics –application to closed and open systems – steady and unsteady flow processes.

UNIT-II Second Law and Entropy

Second law of thermodynamics – Kelvin-Plank’s and Clausius statements. Reversibility and irreversibility. Heat Engine, Refrigerator, Heat pump, Carnot theorem, efficiency, COP.
Clausius inequality, concept of entropy, entropy of ideal gas, principle of increase of entropy.

UNIT-III Properties of Pure Substances

Properties of pure substances - solid, liquid and vapour phases, phase rule, P-V, P-T, T-V, T-S, H-S diagrams, PVT surfaces. Thermodynamic properties of steam, Dryness fraction and its determination by separating and throttling calorimeter.	
UNIT-IV	**Ideal and Real Gases, Gas Mixtures and Thermodynamic Relations**
Properties of ideal and real gases - Equations of state - compressibility factor - compressibility chart. Properties of gas mixture. Exact differentials – Difference and ratio of heat capacities - Clausius Clapeyron equations - Joule Thomson effect.	
UNIT-V	**Psychrometry**
Psychrometric properties, Psychrometric charts. Property calculations of air vapour mixtures by using chart and expressions. Psychrometric process – adiabatic saturation, sensible heating and cooling, humidification, dehumidification, evaporative cooling and adiabatic mixing. Simple Applications.	
Text Books:	
1.Nag.P.K, Engineering Thermodynamics, 6th Edition, Tata McGraw-Hill, 2017. 2. R.K.Rajput, A Textbook of Engineering Thermodynamics, 5 th Edition, Laxmi Publications, 2016. 3. Onkar Singh, Applied Thermodynamics, 4 th Edition, New Age International Private Limited, 2018. 4. Chattopadhyay, P, “Engineering Thermodynamics”, Oxford University Press, 2016.	
Reference Books:	
1. Yunus A Cengel and Michael A Boles, Thermodynamics: An Engineering Approach, 8th Edition, McGraw Hill, 2017. 2. Claus Borgnakke, Richard E. Sonntag, Fundamentals of Thermodynamics, 8th Edition, Wiley, 2016. 3. Lynn D. Russell, George A. Adebiyi, Engineering Thermodynamics, 6th Edition, Oxford University Press, New Delhi, 2008.	